SC4003 CZ/CE4046 Intelligent Agents

Assignment 2

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1 Introduction

The Prisoner's Dilemma is a classical problem in game theory. This assignment extends the repeated Prisoner's Dilemma to a three-player setting, where each round involves three agents simultaneously choosing to either cooperate (0) or defect (1). The payoff for each player depends on the actions of all three, and satisfies the following ordering:

$$U(DCC)>U(CCC)>U(DDC)>U(CDC)>U(DDD)>U(CDD)$$

With the specific payoff values of 8, 6, 5, 3, 2, and 0 respectively for player 1. This preserves the core dilemma structure, where defection always yields higher individual payoff given any response by the other two players, yet full cooperation outperforms full defection. A match consists of approximately 100 rounds, and a player's score is the average payoff across all rounds of that match. Each player can see the full history of all previous actions, thus allowing them to adapt to their opponents' behaviors over the course of a match.

2 Agent Design

CheongEnWei_Ryan_Player is built on a cooperation-first philosophy, seeking to establish and maintain mutually beneficial cooperation when possible, while also having adaptive defection thresholds to protect itself from exploitation by defectors. It operates through three sequential decision phases on each round, evaluated in order of priority.

Phase 1: Early Cooperation (Rounds 0–4)
The agent always cooperates for the first five rounds regardless of opponent behaviour. This signals cooperative intent to establish goodwill, and accumulates a sufficient history window for statistical calculations used for decision making in the later phases.

Phase 2 — Chronic Defector Hard Trigger
From round 5 onwards, the agent scans the full match history to check if there are any two consecutive rounds where both opponents defected together. If such a case is detected, the agent permanently defects for the remainder of the match. This trigger addresses the situation where both opponents are chronic defectors (e.g. two NastyPlayers). One-sided defections are ignored as they might simply be noise or the behavior of a single hostile opponent that the third player would already punish.

Phase 3 — Dynamic Cooperation Threshold

If the chronic defector trigger is not activated, the agent computes the individual cooperation rate for each opponent as:

$$Rate = \frac{Number\ of\ cooperative\ actions\ by\ Opponent}{Number\ of\ rounds}$$

It then compares the rate of each opponent against a dynamic threshold that increases as the match progresses as shown in Table 1.

Round Range	**Rate Threshold**
0 - 69	0.60
70 - 84	0.65
85 - 100	0.72

Table 1: Rate threshold for each round range

If at least one opponent meets or exceeds the threshold, the agent cooperates. Otherwise, it defects. The OR condition is a deliberate design choice motivated by the expectation that the tournament population will consist largely of Tit-for-Tat variants or other cooperation-leaning strategies. In such an environment, it is statistically likely that at least one opponent in any given matchup will favor cooperation. Thus, by requiring only one opponent to meet this threshold rather than both, the agent maximises the number of rounds spent in the high-payoff mutual cooperation state, U(CCC)=6, rather than prematurely triggering defection cycles where the payoff would be low, U(DDD)=2. The increasing threshold accounts for the end-game effect, where rational agents would have an increasing incentive to defect in the later rounds, ensuring the agent is more selective about cooperating as the match nears its end.

3 Experimental Methodology

To evaluate CheongEnWei_Ryan_Player, a Monte Carlo simulation was conducted using a custom tournament environment implemented in Java. Rather than relying on a single tournament run, the simulation was repeated for 1,000 independent iterations to account for the randomness arising from agents such as the RandomPlayer or FreakyPlayer, whose actions involve randomised decisions. Averaging across 1,000 runs ensures that the observed performance is representative of the agent's actual capabilities rather than variance from lucky or unlucky matchups.

In each iteration, all agent types compete against every possible triple combination of players, with each match consisting of approximately 100 rounds. A player's score for a match is the average payoff per round as defined by the payoff matrix in the assignment test script. Two separate tournaments will be held. The first tournament involves CheongEnWei_Ryan_Player and the sample agents provided in the assignment test script. The second tournament uses a curated pool of agents, replacing some baseline agents and adding in custom agents that are designed to reflect a more realistic and diverse competitive population.

Beyond average score per round, the following metrics were collected across all 1,000 iterations to provide a more well-rounded view of each agent's performance:

- **Average score per round:** Mean payoff averaged across all matches and all iterations
- **Top-1 finish rate:** The percentage of iterations in which the agent ranked first overall
- **Top-3 finish rate:** The percentage of iterations in which the agent placed within the top three, used to measure consistency of strong performance
- **Average rank:** The mean finishing position across all iterations, penalising strategies that occasionally score well but frequently finish low

These metrics aid in distinguishing between agents that are reliably strong versus those with high-variance, which is important in a tournament where a consistent mid-to-high placement accumulates a better overall score than a strategy that performs erratically.

4 Evaluation Tournament 1: Baseline Agents

Table 2 presents the results of the first tournament, in which CheongEnWei_Ryan_Player competed against the six sample agents provided in the assignment test script, averaged across 1,000 independent simulation runs.

Agent	Avg Score/Round	Top-1 Rate	Top-3 Rate	Avg Rank
CheongEnWei_ Ryan_Player	4.71	64.8%	99.7%	1.47
Tolerant Player	4.57	18.1%	94.0%	2.30
T4TPlayer	4.55	17.0%	94.2%	2.38
NicePlayer	4.21	0.1%	8.4%	4.36
FreakyPlayer	4.08	0.0%	3.6%	5.09
NastyPlayer	3.94	0.0%	0.1%	5.81
RandomPlayer	3.77	0.0%	0.0%	6.58

Table 2: Tournament 1 results averaged over 1,000 Monte Carlo iterations.

CheongEnWei_Ryan_Player ranked first overall across all four metrics, achieving the highest average score of 4.71 points per round , 2.9% higher than the next best agent, TolerantPlayer (4.57). Notably, it secured a Top-1 finish 64.8% of the time and a Top-3 finish 99.7% of the time, demonstrating that its strong performance is consistent rather than due to lucky rounds. Its average rank of 1.47 further confirms this, placing it close to first place on average across all 1,000 tournament iterations.

TolerantPlayer and T4TPlayers, the two closest competitors, performed similarly to each other with very similar average scores and Top-3 rates. This is expected as both are cooperative history-aware strategies that tend to achieve better scores in populations where most agents tend to cooperate. CheongEnWei_Ryan_Player's edge over these agents stems from two unique properties. First, its chronic defector trigger only activates if both opponents defect together for two consecutive rounds, making it more forgiving than the T4TPlayer. This reduces unnecessary defection cycles in rounds where only one opponent is hostile. Second, its dynamic threshold mechanism is more selective, offering better protection against defection than TolerantPlayer. Together, these properties allow CheongEnWei_Ryan_Player to capture the benefits of the TolerantPlayer and T4TPlayer’s approaches.

NicePlayer, despite always cooperating, placed a distant fourth as it gets exploited by hostile agents like NastyPlayer and FreakyPlayer. NastyPlayer itself scores poorly (3.94) as it tends to cause a mutual defection across most matchups, yielding a per round payoff of only 2. RandomPlayer finishes last as its unpredictable behavior prevents it from maintaining a cooperative equilibrium.

5 Evaluation Tournament 2: Curated Agents

Three of the agents in Tournament 2 are custom-designed strategies not found in the assignment test script. A brief description of each is provided below.

- **PavlovPlayer:** A Win-Stay-Lose-Shift strategy. The agent repeats its previous action if the resulting payoff was satisfactory (≥ 5), and switches otherwise. It is highly adaptive to immediate outcomes but vulnerable to sustained defection pressure from multiple opponents simultaneously.
- **TFTTPlayer (Tit-for-Two-Tats):** A more forgiving variant of TFT that only retaliates after an opponent defects in two consecutive rounds, rather than immediately. This tolerance makes it more resistant to noise or isolated defections, but susceptible to opponents who defect in alternating rounds.
- **HitAndRunPlayer:** A periodic defector that cooperates by default but injects a deliberate defection every 7 rounds, followed by two rounds of cooperation to avoid immediate retaliation. It otherwise mirrors TFT behaviour. This strategy is designed to extract small periodic gains from cooperative partners while limiting punishment and retaliation.

Table 3 presents the results of the second tournament, in which CheongEnWei_Ryan_Player competed against 11 baseline and curated agents representing a broader range of strategies, averaged across 1,000 independent simulation runs.

Agent	Avg Score/Round	Top-1 Rate	Top-3 Rate	Avg Rank
CheongEnWei_Ryan_Player	5.20	91.1%	100.0%	1.10
TolerantPlayer	5.19	6.9%	100.0%	2.23
T4TTPlayer	5.19	2.0%	100.0%	2.67
T4TPlayer	5.12	0.0%	0.0%	4.00

NicePlayer	4.88	0.0%	0.0%	5.00
PavlovPlayer	4.79	0.0%	0.0%	6.26
HitAndRunPlayer	4.78	0.0%	0.0%	6.74
NastyPlayer	3.48	0.0%	0.0%	8.00

Table 3: Tournament 2 results averaged over 1,000 Monte Carlo iterations.

CheongEnWei_Ryan_Player placed first overall across all four metrics, achieving the highest average score of 5.20 and a dominant Top-1 finish rate of 91.1%. Crucially, it placed within the top three in every single iteration, yielding a perfect Top-3 rate of 100% and an average rank of 1.10, indicating it finished first in the overwhelming majority of runs. Its two closest competitors, TolerantPlayer (5.1925) and TFTTPlayer (5.1914), trailed by a margin of just 0.003 points per round. Both also achieved 100% Top-3 rates, confirming that the top three positions are consistently occupied by cooperative history-aware strategies in this population. But CheongEnWei_Ryan_Player has a slight edge compared to these two agents as it more reliably converts strong performance into first-place finishes, as seen in its high Top-1 finish rate.

The performance gap between the top three and the rest of the field is meaningful. T4TPlayer, despite being a well-established strategy, scores noticeably lower as it mirrors only one randomly chosen opponent per round, occasionally failing to respond appropriately to the other, reducing its chances of forming a consistent mutual cooperation. NicePlayer gets exploited by hostile agents, pulling its scores down. PavlovPlayer and HitAndRunPlayer score similarly as their defections cause mutual defection cycles against the Tit-For-Tat variant majority. NastyPlayer finishes last as its constant defection causes mutual defection cycles against most opponents, which is consistent with its performance in Tournament 1.

6 Conclusion

This report presented the design and evaluation of CheongEnWei_Ryan_Player for the three-player Repeated Prisoner's Dilemma. Evaluation across two Monte Carlo tournaments of 1,000 iterations each demonstrated consistent top-tier performance. These results validate the core design assumption that a cooperative population, reflective of the expected student tournament environment, rewards agents that sustain mutual cooperation while being able to defend itself against exploitation.